How Brains Learn

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: conditioning food sound stronger weight wire

You can use a word more than once.

1. Your other hand. Write your name with your other hand. Your partner times you.

	Try 1	Try 2	Try 3
Time (seconds)			

2. Stronger connections.

When you practice, the synapses between your neurons get _____.

“Neurons that fire together, _____ together.”

3. Pavlov’s dogs.

Pavlov’s team played a _____ right before they gave the dogs _____.

Soon the dogs drooled when they heard the sound, even with no _____.

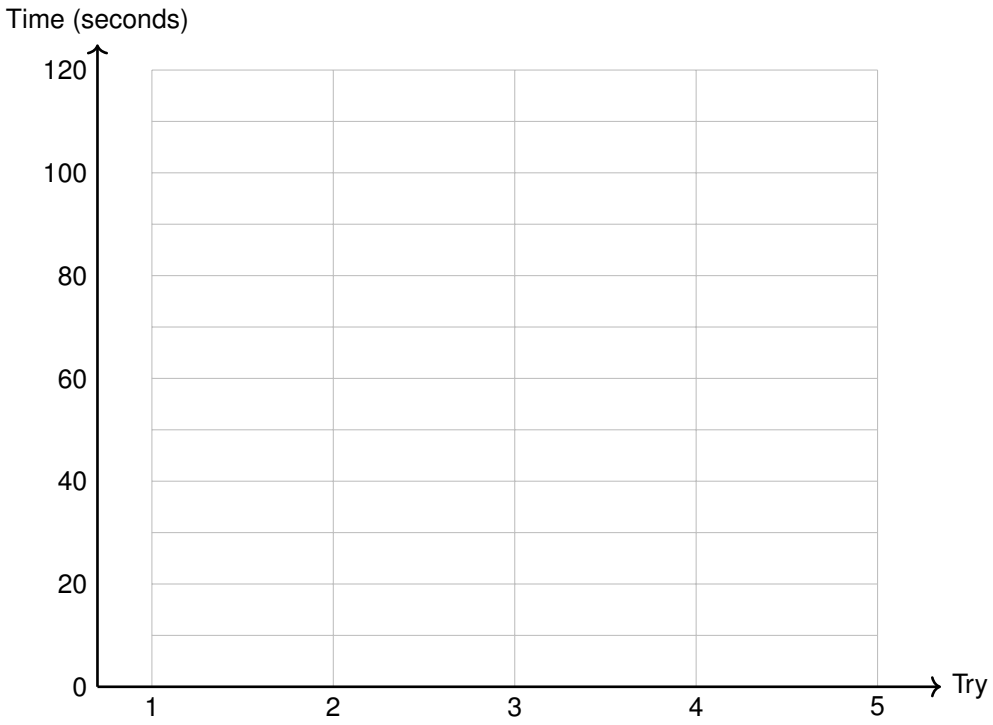
Learning to connect two things that happen together is called _____.

In AI, the strength of a connection is a number called a _____.

4. Mirror drawing. Use the same pen for all 5 tries.

	Try 1	Try 2	Try 3	Try 4	Try 5
Time (seconds)					
Mistakes					

5. My learning curve. Put a dot for each try. Connect the dots.



What happened to your time? _____

6. Talk about it. Your brain changes synapses. AI changes weights. Is it the same kind of learning?

Brain learning and AI learning are **the same** / **different** because _____

_____

A sound that makes me reach for my phone: _____ That is like Pavlov's dogs.

_____

Go Further (Day 2). A new mirror: left and right flipped.

Same pen as yesterday: _____ seconds. New mirror: _____ _____ _____ seconds.

Was it hard again? _____ Why? _____